

Procedural Programming in C#

Overview: Procedural Programming is a programming paradigm that relies on a structured sequence of instructions, organized into procedures, functions, or routines to perform tasks. It follows a top-down approach and emphasizes a clear sequence of commands and control flow.

Procedural programming (PP) follows a step-by-step approach, where programs are divided into functions.

Examples: C, Pascal, C#

Focus: Functions or procedures to perform operations.

Sample Program

Write a C# program to calculate the total and average marks of a student based on marks in three subjects using a procedural programming approach.

```
Java
using System;
class StudentMarks
{
    static void Main()
    {
        // Taking input for three subjects
        Console.Write("Enter marks for Subject 1: ");
        int subject1 = Convert.ToInt32(Console.ReadLine());

        Console.Write("Enter marks for Subject 2: ");
        int subject2 = Convert.ToInt32(Console.ReadLine());

        Console.Write("Enter marks for Subject 3: ");
        int subject3 = Convert.ToInt32(Console.ReadLine());

        // Calculating total and average using functions
        int totalMarks = CalculateTotal(subject1, subject2, subject3);
        double averageMarks = CalculateAverage(totalMarks, 3);

        // Displaying results
        Console.WriteLine("Total Marks: " + totalMarks);
        Console.WriteLine("Average Marks: " + averageMarks);
    }

    // Function to calculate total marks
    static int CalculateTotal(int s1, int s2, int s3)
    {
        return s1 + s2 + s3;
    }
}
```

```
}  
// Function to calculate average marks  
static double CalculateAverage(int total, int subjects)  
{  
    return (double)total / subjects;  
}  
}
```

Output Example:

Input:

Enter the student's name: Alice

Enter marks for Subject 1: 85

Enter marks for Subject 2: 90

Enter marks for Subject 3: 88

Output:

Student Name: Alice

Total Marks: 263

Average Marks: 87.67

Explanation:

- The program prompts the user to enter marks for three subjects.
- The CalculateTotal function computes the sum of marks.
- The CalculateAverage function computes the average by dividing the total marks by the number of subjects.
- The DisplayResults function prints the results.
- The results are displayed in the console output.

This follows a **procedural programming** approach by dividing the logic into functions for better readability and reusability.

2. Object-Oriented Programming (OOP) Overview An Object-Oriented Programming (OOP) Language is a programming paradigm based on the concept of "objects", which are instances of classes. It allows developers to model real-world entities and solve problems by organizing data (attributes) and behaviors (methods) into objects.

The OOP paradigm organizes code into classes and objects, focusing on data (attributes) and behaviors (methods).

Advantages of OOP Over Procedural Programming

- **Encapsulation:** Combines data and methods into a single unit (class) and restricts direct access to certain components, ensuring data security.
- **Abstraction:** Hides complex implementation details, exposing only essential features.
- **Reusability Through Inheritance:** Enables classes to inherit properties and behaviors from other classes.
- **Polymorphism:** Allows objects to take multiple forms, enabling method overloading and overriding.
- **Natural Modeling of Real-World Entities:** OOP maps objects in the real world to programming constructs.
- **Modularity and Maintainability:** Divides the program into small, manageable pieces (classes).

Comparison Between Procedural Programming and OOP

Feature	Procedural Programming	Object-Oriented Programming
Focus	Functions and procedures.	Classes and objects.
Data Security	Global data is prone to accidental modification.	Encapsulated, secure data access.
Reusability	Minimal reusability, and code duplication.	Reusable code through inheritance and polymorphism.
Scalability	Poor scalability for large systems.	Highly scalable and modular.
Real-World Mapping	Difficult to map real-world entities.	Naturally models real-world objects.
Maintenance	Difficult to maintain in large projects.	Easier to maintain with a modular design.